ELITE IGCSE ACADEMY

Student Past Paper Solutions

May 2025 4WM1H Worked Solutions

May2025 | 4WM1H | Unit 1

31 questions ▪ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 3

TOPIC: **Fractions**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

1 Without using a calculator, work out

$$3\frac{2}{3} \times 2\frac{4}{7}$$

Show each stage of your working.
Give your answer as a mixed number in its simplest form.

....

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Fractions**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Was tagged Algebra Toolkit; actually Fractions.

Solution

Convert the mixed numbers to improper fractions.

$$3\frac{2}{3} = \frac{11}{3}, \qquad 2\frac{4}{7} = \frac{18}{7}$$

Then multiply.

$$\frac{11}{3} \times \frac{18}{7} = \frac{198}{21} = \frac{66}{7}$$

Convert to a mixed number.

$$\frac{66}{7} = 9\frac{3}{7}$$

FINAL ANSWER

$9\frac{3}{7}$

PAST PAPER QUESTION

Q#: 2 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

2 The height of a tree is 17.8 metres correct to one decimal place.

(a) Write down the lower bound for the height of the tree.

.... metres

(1)

The weight of an apple is 145 grams correct to the nearest 5 grams.

(b) Write down the upper bound for the weight of the apple.

.... grams

(1)

$$A = \frac{4.766 \times 815}{0.399}$$

(c) By rounding each number to one significant figure, work out an estimate for the value of A

Show your working clearly.

....

(2)

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Rounding, estimation and bounds. The tag is correct.

Solution

For 17.8 correct to 1 decimal place, the lower bound is

$$17.8 - 0.05 = 17.75$$

For 145 correct to the nearest 5 grams, the upper bound is

$$145 + 2.5 = 147.5$$

For the estimate,

$$A = \frac{4.766 \times 815}{0.399}$$

Round to 1 significant figure.

$$4.766 \approx 5, \quad 815 \approx 800, \quad 0.399 \approx 0.4$$

$$A \approx \frac{5 \times 800}{0.4} = 10000$$

FINAL ANSWER

17.75 m, 147.5 g, and 10000.

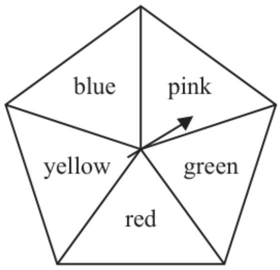
PAST PAPER QUESTION

TOPIC: **Probability Toolkit**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

Q#: 3 Marks: 4

3 Here is a biased spinner.
The spinner can land on pink or on green or on red or on yellow or on blue.



The table gives information about the probability that, when the spinner is spun, it will land on each colour.

Colour	pink	green	red	yellow	blue
Probability	0.14	0.23	$2x$	0.18	$7x$

Jim is going to spin the spinner 500 times.

Work out the expected number of times the spinner will land on blue.

....

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 4

TOPIC: **Probability Toolkit**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Probability Toolkit. The tag is correct.

Solution

The probabilities add to 1.

$$0.14 + 0.23 + 2x + 0.18 + 7x = 1$$

$$0.55 + 9x = 1$$

$$x = 0.05$$

The probability of blue is

$$7x = 7(0.05) = 0.35$$

Expected number of blue results:

$$500 \times 0.35 = 175$$

FINAL ANSWER

175

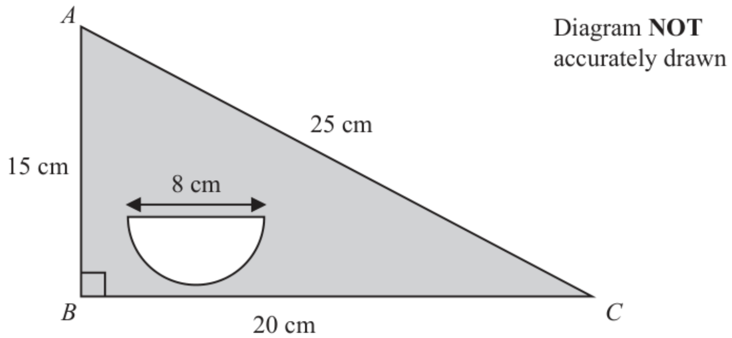
PAST PAPER QUESTION

Q#: 4 Marks: 3

TOPIC: **Area & Perimeter**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

4 The diagram shows triangle ABC and a semicircle.



$AB = 15$ cm $\quad BC = 20$ cm $\quad AC = 25$ cm $\quad$ angle $ABC = 90°$

The diameter of the semicircle is 8 cm

Work out the area of the region shown shaded in the diagram.
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 4 is 3 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 3

TOPIC: **Area & Perimeter**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

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Topic check. Perimeter and area. The tag is correct.

Solution

The shaded area is the triangle area minus the semicircle area.

$$\text{triangle area} = \frac{1}{2} \times 15 \times 20 = 150$$

The semicircle has diameter 8, so radius 4.

$$\text{semicircle area} = \frac{1}{2}\pi(4)^2 = 8\pi$$

$$\text{shaded area} = 150 - 8\pi = 124.867\ldots$$

To 3 significant figures,

$$125$$

FINAL ANSWER

125 cm^2

PAST PAPER QUESTION Q#: 5 Marks: 4

TOPIC: **Algebraic Roots & Indices**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

5 $x^5 \times x^7 = x^m$

(a) Find the value of m

$m =$ ....
(1)

$y^8 \div y^3 = y^n$

(b) Find the value of n

$n =$ ....
(1)

(c) Simplify fully $(5a^4r^2)^3$

....
(2)

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Algebraic Roots & Indices**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Was tagged Algebra Toolkit; actually Algebraic Roots & Indices.

Solution

$$x^5 \times x^7 = x^{5+7} = x^{12}$$

So $m = 12$.

$$y^8 \div y^3 = y^{8-3} = y^5$$

So $n = 5$.

$$(5a^4r^2)^3 = 5^3(a^4)^3(r^2)^3 = 125a^{12}r^6$$

FINAL ANSWER

$m = 12$, $n = 5$, and $125a^{12}r^6$.

PAST PAPER QUESTION

Q#: 6 Marks: 3

TOPIC: **Standard & Compound Units**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

6 The diagram shows a block in the shape of a cuboid.

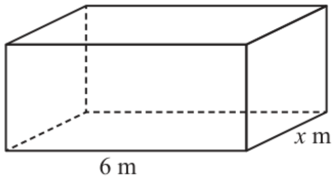

Diagram **NOT** accurately drawn

The block has a horizontal base with length 6 m and width x m

The block is placed on a table so that the whole of its horizontal base is in contact with the table.

The block exerts a force of 702 newtons on the table.
The pressure on the table due to the block is 65 newtons/m²

Work out the value of x

$$\boxed{\text{pressure} = \frac{\text{force}}{\text{area}}}$$

x = ...

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 3

TOPIC: **Standard & Compound Units**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Standard and compound units. The tag is correct.

Solution

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$65 = \frac{702}{\text{area}}$$

$$\text{area} = \frac{702}{65} = 10.8$$

The base area is $6x$, so

$$6x = 10.8$$

$$x = 1.8$$

FINAL ANSWER

$x = 1.8$

PAST PAPER QUESTION

Q#: 7 Marks: 6

TOPIC: **Factorising**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

7 (a) Solve $x - 4 = \dfrac{3 + 2x}{6}$

Show clear algebraic working.

$x =$ ....

(3)

(b) (i) Factorise $y^2 - 11y + 30$

....

(2)

(ii) Hence solve $y^2 - 11y + 30 = 0$

....

(1)

(Total for Question 7 is 6 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 6

TOPIC: **Factorising**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Factorising. The tag is correct.

Solution

For part (a),

$$x - 4 = \frac{3 + 2x}{6}$$

Multiply by 6.

$$6x - 24 = 3 + 2x$$

$$4x = 27$$

$$x = \frac{27}{4} = 6.75$$

For part (b),

$$y^2 - 11y + 30 = (y - 5)(y - 6)$$

Hence,

$$(y - 5)(y - 6) = 0$$

$$y = 5 \quad \text{or} \quad y = 6$$

FINAL ANSWER

$x = 6.75$, $(y - 5)(y - 6)$, and $y = 5$ or 6.

PAST PAPER QUESTION

Q#: 8 Marks: 5

TOPIC: **Set Notation & Venn Diagrams**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

8 $\mathscr{E} = \{18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30\}$

$A = \{\text{even numbers}\}$

$B = \{\text{multiples of 3}\}$

$C = \{\text{multiples of 5}\}$

(a) List the members of the set $(A \cup B)'$

....

(2)

Sophie writes down the statement $B \cap C = \varnothing$

(b) Explain why Sophie's statement is wrong.

....

....

(1)

D is a set such that $A \cap D = \{18, 26\}$
The set D has exactly 5 members.

(c) List the members of one possible set D

....

(2)

(Total for Question 8 is 5 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 5

TOPIC: **Set Notation & Venn Diagrams**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Was tagged Algebra Toolkit; actually Set Notation & Venn Diagrams.

Solution

$$A = \{18, 20, 22, 24, 26, 28, 30\}$$

$$B = \{18, 21, 24, 27, 30\}$$

$$A \cup B = \{18, 20, 21, 22, 24, 26, 27, 28, 30\}$$

So

$$(A \cup B)' = \{19, 23, 25, 29\}$$

Sophie is wrong because 30 is both a multiple of 3 and a multiple of 5, so $30 \in B \cap C$.
For D, include 18 and 26, but no other even numbers. One possible set is

$$D = \{18, 19, 23, 25, 26\}$$

FINAL ANSWER
$\{19, 23, 25, 29\}$; Sophie is wrong because $30 \in B \cap C$; one possible D is $\{18, 19, 23, 25, 26\}$.

PAST PAPER QUESTION

Q#: 9 **Marks: 3**

TOPIC: **Algebraic Roots & Indices**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

9 $7^x = 1$

(a) Write down the value of x

$x =$ ....

(1)

$$\frac{8^{-4} \times 8^{11}}{8^{12}} = 8^n$$

(b) Work out the value of n

$n =$ ....

(2)

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Algebraic Roots & Indices**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Was tagged Algebra Toolkit; actually Algebraic Roots & Indices.

Solution

$$7^x = 1$$

Since $7^0 = 1$,

$$x = 0$$

For part (b),

$$\frac{8^{-4} \times 8^{11}}{8^{12}} = 8^n$$

$$8^{-4} \times 8^{11} = 8^7$$

$$\frac{8^7}{8^{12}} = 8^{7-12} = 8^{-5}$$

So $n = -5$.

FINAL ANSWER

$x = 0$, $n = -5$.

PAST PAPER QUESTION Q#: 10 Marks: 5

TOPIC: **Area & Perimeter**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

10

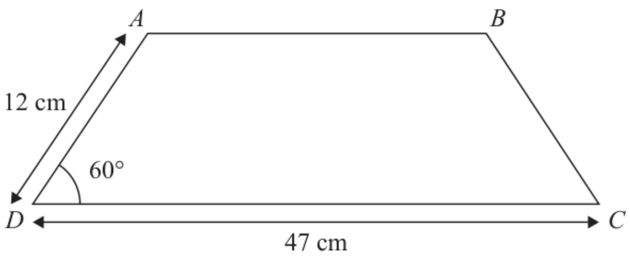

Diagram **NOT** accurately drawn

ABCD is a trapezium with one line of symmetry.

angle $ADC = 60°$ $AD = 12$ cm $DC = 47$ cm

Work out the area of the trapezium.
Give your answer correct to 3 significant figures.
Show your working clearly.

.... cm^2

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 5

TOPIC: **Area & Perimeter**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Perimeter and area. The tag is correct.

Solution

The trapezium is symmetrical. The horizontal shift from each sloping side is

$$12\cos 60^\circ = 6$$

So the top parallel side is

$$47 - 6 - 6 = 35$$

The height is

$$12\sin 60^\circ = 10.392\ldots$$

Area of the trapezium:

$$\frac{1}{2}(47+35)(12\sin 60^\circ) = 426.084\ldots$$

To 3 significant figures,

$$426$$

FINAL ANSWER

426 cm^2

PAST PAPER QUESTION

Q#: 11 Marks: 6

TOPIC: **Expanding Brackets**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

11 (a) Expand and simplify $5x(5x + 4)(2x - 3)$

....

(3)

(b) Express $\dfrac{9}{4} + \dfrac{y - 7}{5y}$ as a single fraction in its simplest form.

....

(3)

(Total for Question 11 is 6 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 6

TOPIC: **Expanding Brackets**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Expanding brackets. The tag is correct.

Solution

$$5x(5x+4)(2x-3)$$

First,

$$(5x+4)(2x-3) = 10x^2 - 15x + 8x - 12 = 10x^2 - 7x - 12$$

Then multiply by $5x$.

$$5x(10x^2 - 7x - 12) = 50x^3 - 35x^2 - 60x$$

For part (b), use denominator $20y$.

$$\frac{9}{4} + \frac{y-7}{5y} = \frac{45y}{20y} + \frac{4(y-7)}{20y}$$

$$= \frac{49y-28}{20y}$$

FINAL ANSWER

$50x^3 - 35x^2 - 60x$ and $\frac{49y-28}{20y}$.

PAST PAPER QUESTION Q#: 12 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

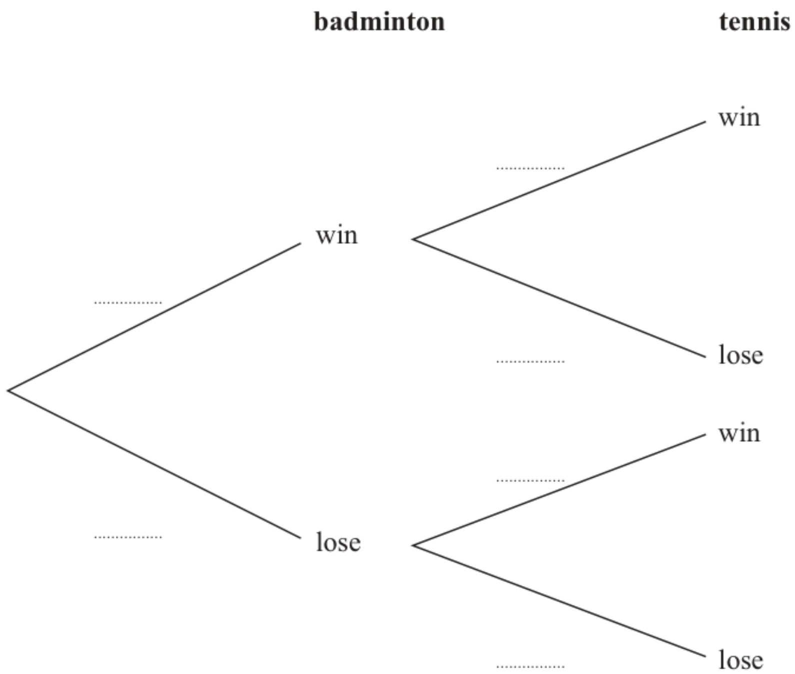
May 2025 4WM1H - May2025 | 4WM1H | Unit 1

12 Ricardo is going to play one game of badminton and one game of tennis.

He can either win or lose each game.

The probability that he will win the game of badminton is 0.7
The probability that he will win the game of tennis is 0.4

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Ricardo loses both games.

....

(2)

(Total for Question 12 is 4 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Probability tree diagrams. The tag is correct.

Solution

For badminton,

$$P(\text{win}) = 0.7, \qquad P(\text{lose}) = 0.3$$

For tennis,

$$P(\text{win}) = 0.4, \qquad P(\text{lose}) = 0.6$$

The probability of losing both games is

$$0.3 \times 0.6 = 0.18$$

FINAL ANSWER

Tennis branches are 0.4 and 0.6 on each branch; probability of losing both is 0.18.

PAST PAPER QUESTION **Q#: 13** **Marks: 3**

TOPIC: **Graphs of Functions**

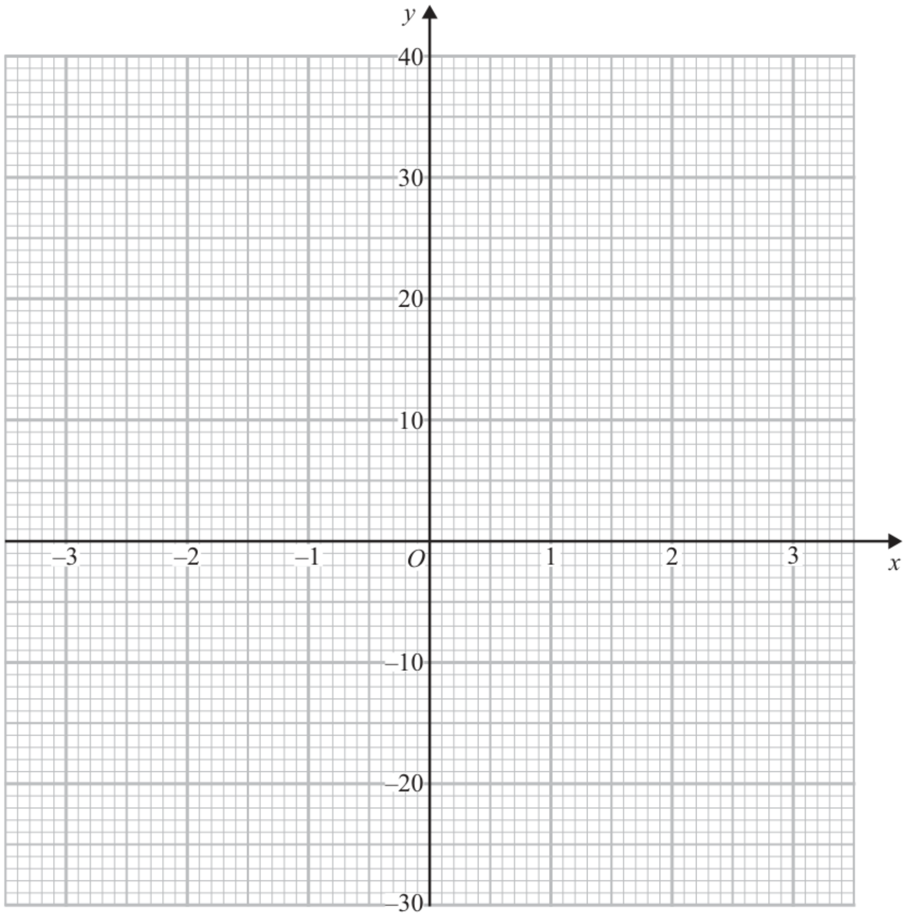
May 2025 4WM1H - May2025 | 4WM1H | Unit 1

13 (a) Complete the table of values for $y = x^3 + 2x + 3$

x	-3	-2	-1	0	1	2	3
y		-9	0	3	6		36

(1)

(b) On the grid, draw the graph of $y = x^3 + 2x + 3$ for $-3 \leqslant x \leqslant 3$



(2)

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 3

TOPIC: **Graphs of Functions**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Graphs of functions. The tag is correct.

Solution

Use $y = x^3 + 2x + 3$.
For $x = -3$,

$$y = (-3)^3 + 2(-3) + 3 = -30$$

For $x = 2$,

$$y = 2^3 + 2(2) + 3 = 15$$

So the table is

x	-3	-2	-1	0	1	2	3
y	-30	-9	0	3	6	15	36

Plot these points and draw a smooth cubic curve through them.

FINAL ANSWER
Missing values -30 and 15; draw the smooth curve through the table points.

PAST PAPER QUESTION **Q#: 14** **Marks: 4**

TOPIC: **Circles, Arcs & Sectors**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

14 The diagram shows sector OAB of a circle, centre O

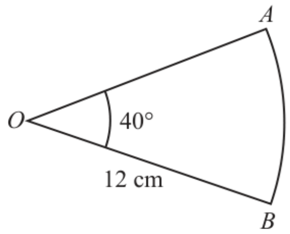

Diagram **NOT** accurately drawn

$OA = OB = 12$ cm angle $AOB = 40°$

(a) Work out the area of the sector.
Give your answer correct to 3 significant figures.

.... cm^2
(2)

The diagram shows triangle XYZ

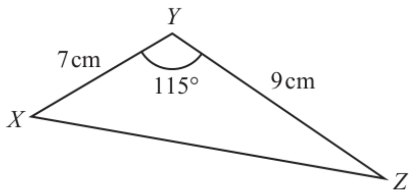

Diagram **NOT** accurately drawn

$XY = 7$ cm $YZ = 9$ cm angle $XYZ = 115°$

(b) Work out the area of the triangle.
Give your answer correct to 3 significant figures.

.... cm^2
(2)

(Total for Question 14 is 4 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 4

TOPIC: **Circles, Arcs & Sectors**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Circles, arcs and sectors. The tag is correct.

Solution

Sector area:

$$\frac{40}{360}\pi(12)^2 = 16\pi = 50.265\ldots$$

To 3 significant figures,

$$50.3 \text{ cm}^2$$

Triangle area:

$$\frac{1}{2}(7)(9)\sin 115^\circ = 28.548\ldots$$

To 3 significant figures,

$$28.5 \text{ cm}^2$$

FINAL ANSWER

50.3 cm^2 and 28.5 cm^2.

PAST PAPER QUESTION

Q#: 15 **Marks: 3**

TOPIC: **Solving Quadratic Equations**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

15 Solve the equation $5x^2 + 9x - 17 = 0$
Show your working clearly.
Give your solutions correct to 3 significant figures.

....

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 3

TOPIC: **Solving Quadratic Equations**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Was tagged Solving Linear Equations; actually Quadratic Formula.

Solution

$$5x^2 + 9x - 17 = 0$$

Use the quadratic formula with $a = 5$, $b = 9$, $c = -17$.

$$x = \frac{-9 \pm \sqrt{9^2 - 4(5)(-17)}}{2(5)}$$

$$x = \frac{-9 \pm \sqrt{421}}{10}$$

$$x = 1.152\ldots \quad \text{or} \quad x = -2.952\ldots$$

To 3 significant figures,

$$x = 1.15 \quad \text{or} \quad x = -2.95$$

FINAL ANSWER

$x = 1.15$ or $x = -2.95$.

PAST PAPER QUESTION

Q#: 16 **Marks: 3**

TOPIC: **Algebraic Roots & Indices**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

16 Solve the equation $8^{3y+4} \times 4^{3y} = 2^{5y}$
Show your working clearly.

$y = $ ....

(Total for Question 16 is 3 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 3

TOPIC: **Algebraic Roots & Indices**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Was tagged Solving Linear Equations; actually Algebraic Roots & Indices.

Solution

$$8^{3y+4} \times 4^{3y} = 2^{5y}$$

Write 8 and 4 as powers of 2.

$$(2^3)^{3y+4}(2^2)^{3y} = 2^{5y}$$

$$2^{9y+12} \times 2^{6y} = 2^{5y}$$

$$2^{15y+12} = 2^{5y}$$

So

$$15y + 12 = 5y$$

$$y = -\frac{6}{5}$$

FINAL ANSWER

$y = -\frac{6}{5}$

PAST PAPER QUESTION Q#: 17 Marks: 4

TOPIC: **Surds**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

17 $\left(\sqrt{3}\right)^5 = k\sqrt{3}$ where k is an integer.

(a) Find the value of k

$k =$ ....

(1)

(b) Show that $\dfrac{21}{3-\sqrt{2}}$ can be written in the form $c + \sqrt{d}$

where c and d are integers.
Show each stage of your working clearly.

(3)

(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 4

TOPIC: **Surds**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Was tagged Algebra Toolkit; actually Surds.

Solution

$$(\sqrt{3})^5 = (\sqrt{3})^4\sqrt{3} = 3^2\sqrt{3} = 9\sqrt{3}$$

So $k = 9$.
For part (b), rationalise the denominator.

$$\frac{21}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}} = \frac{21(3+\sqrt{2})}{9-2}$$

$$= 3(3+\sqrt{2}) = 9 + 3\sqrt{2}$$

Since $3\sqrt{2} = \sqrt{18}$,

$$9 + 3\sqrt{2} = 9 + \sqrt{18}$$

FINAL ANSWER
$k = 9$, and $9 + \sqrt{18}$.

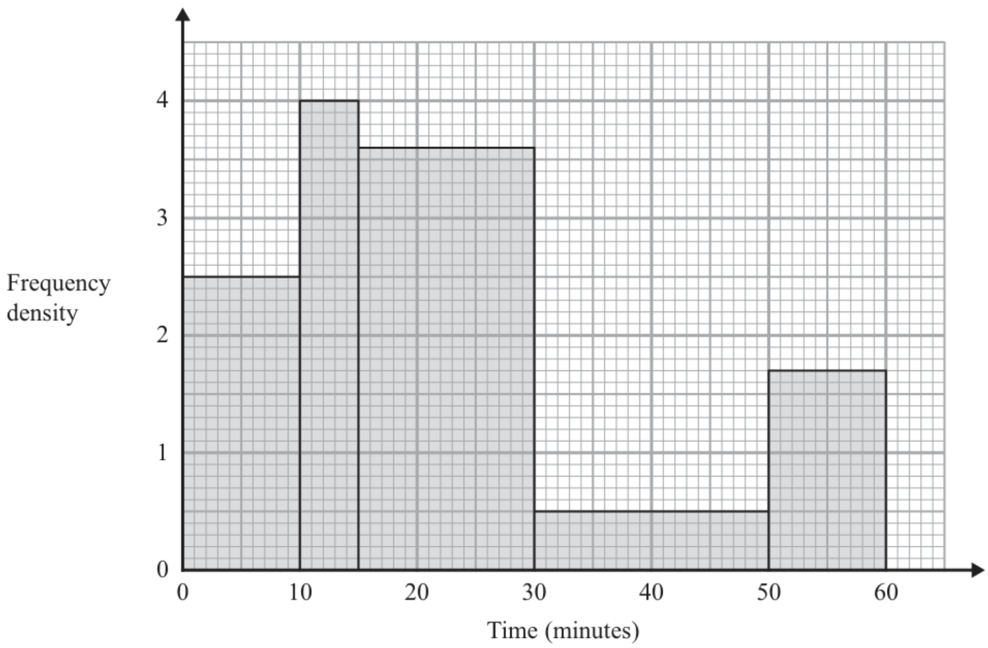
PAST PAPER QUESTION

Q#: 18 Marks: 3

TOPIC: **Histograms**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

18 The histogram shows information about the times, in minutes, that some people were in a shop.



Work out an estimate for the proportion of these people who were in the shop for more than 40 minutes.

....

(Total for Question 18 is 3 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 3

Topic: **Histograms**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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Topic check. Was tagged Standard & Compound Units; actually Histograms.

Solution

Frequency is class width multiplied by frequency density.

$$0\text{–}10 : 10 \times 2.5 = 25$$

$$10\text{–}15 : 5 \times 4 = 20$$

$$15\text{–}30 : 15 \times 3.6 = 54$$

$$30\text{–}50 : 20 \times 0.5 = 10$$

$$50\text{–}60 : 10 \times 1.7 = 17$$

Total:

$$25 + 20 + 54 + 10 + 17 = 126$$

More than 40 minutes is the part from 40 to 50, plus 50 to 60.

$$10 \times 0.5 + 17 = 22$$

So the proportion is

$$\frac{22}{126} = \frac{11}{63}$$

FINAL ANSWER

$\frac{11}{63}$

PAST PAPER QUESTION **Q#: 19** **Marks: 3**

TOPIC: **Algebraic Fractions**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

19 Simplify fully $\dfrac{3x^2 - 3y^2}{5x + 5y} \div \dfrac{xy^2 - x^2y}{10xy}$

Show clear algebraic working.

....

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 3

TOPIC: **Algebraic Fractions**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Was tagged Algebra Toolkit; actually Algebraic Fractions.

Solution

$$\frac{3x^2 - 3y^2}{5x + 5y} \div \frac{xy^2 - x^2y}{10xy}$$

Factorise.

$$= \frac{3(x-y)(x+y)}{5(x+y)} \div \frac{xy(y-x)}{10xy}$$

$$= \frac{3(x-y)}{5} \div \frac{y-x}{10}$$

$$= \frac{3(x-y)}{5} \times \frac{10}{y-x}$$

Since $y - x = -(x - y)$, the expression is

$$-6$$

FINAL ANSWER

-6

PAST PAPER QUESTION **Q#: 20** **Marks: 3**

TOPIC: **Algebraic Fractions**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

20 Given that $k = m + n$ and $m = \dfrac{3}{4n}$

express $\dfrac{7k}{5 - m}$ in the form $\dfrac{a + bn^2}{cn - d}$ where a, b and c are integers and d is prime.

....

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **Algebraic Fractions**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Was tagged Algebra Toolkit; actually Algebraic Fractions.

Solution

$$k = m + n, \qquad m = \frac{3}{4n}$$

So

$$k = \frac{3}{4n} + n = \frac{3 + 4n^2}{4n}$$

Then

$$7k = \frac{21 + 28n^2}{4n}$$

and

$$5 - m = 5 - \frac{3}{4n} = \frac{20n - 3}{4n}$$

Therefore

$$\frac{7k}{5 - m} = \frac{\frac{21+28n^2}{4n}}{\frac{20n-3}{4n}}$$
$$= \frac{21 + 28n^2}{20n - 3}$$

FINAL ANSWER

$\frac{21+28n^2}{20n-3}$

PAST PAPER QUESTION

Q#: 20 Marks: 3

TOPIC: **Algebraic Fractions**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

20 Given that $k = m + n$ and $m = \dfrac{3}{4n}$

express $\dfrac{7k}{5 - m}$ in the form $\dfrac{a + bn^2}{cn - d}$ where a, b and c are integers and d is prime.

....

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **Algebraic Fractions**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Was tagged Algebra Toolkit; actually Algebraic Fractions.

Solution

$$k = m + n, \qquad m = \frac{3}{4n}$$

So

$$k = \frac{3}{4n} + n = \frac{3 + 4n^2}{4n}$$

Then

$$7k = \frac{21 + 28n^2}{4n}$$

and

$$5 - m = 5 - \frac{3}{4n} = \frac{20n - 3}{4n}$$

Therefore

$$\frac{7k}{5 - m} = \frac{\frac{21+28n^2}{4n}}{\frac{20n-3}{4n}}$$

$$= \frac{21 + 28n^2}{20n - 3}$$

FINAL ANSWER

$\frac{21+28n^2}{20n-3}$

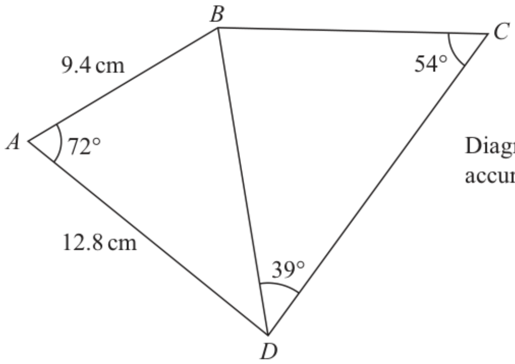
PAST PAPER QUESTION

Q#: 21 Marks: 5

Topic: **Sine, Cosine Rule & Area of Triangles**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

21



Work out the length of *BC*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Was tagged Algebra Toolkit; actually Sine & Cosine Rules.

Solution

First find BD in triangle ABD using the cosine rule.

$$BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8)\cos 72^\circ$$

$$BD = 13.3356\ldots$$

In triangle BCD, use the sine rule.

$$\frac{BC}{\sin 39^\circ} = \frac{BD}{\sin 54^\circ}$$

$$BC = 13.3356\ldots \times \frac{\sin 39^\circ}{\sin 54^\circ}$$

$$BC = 10.3735\ldots$$

To 3 significant figures,

$$BC = 10.4$$

FINAL ANSWER

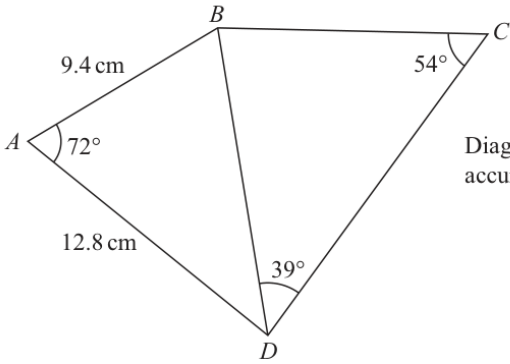
10.4 cm

PAST PAPER QUESTION **Q#: 21** **Marks: 5**

TOPIC: **Sine, Cosine Rule & Area of Triangles**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

21



Work out the length of *BC*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

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To 3 significant figures,

$$BC = 10.4$$

FINAL ANSWER

10.4 cm

PAST PAPER QUESTION

Q#: 22 Marks: 3

TOPIC: **Probability Toolkit**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

22 A box contains 20 counters.

9 of the counters are red
7 of the counters are yellow
4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

....

(Total for Question 22 is 3 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 3

TOPIC: **Probability Toolkit**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Probability Toolkit. The tag is correct.

Solution

Total ways to choose 3 counters:

$$\binom{20}{3} = 1140$$

Exactly two red:

$$\binom{9}{2}\binom{11}{1} = 396$$

Exactly two yellow:

$$\binom{7}{2}\binom{13}{1} = 273$$

Exactly two green:

$$\binom{4}{2}\binom{16}{1} = 96$$

Favourable outcomes:

$$396 + 273 + 96 = 765$$

$$P = \frac{765}{1140} = \frac{51}{76}$$

FINAL ANSWER

$\frac{51}{76}$

PAST PAPER QUESTION Q#: 22 Marks: 3

TOPIC: **Probability Toolkit**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

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....

(Total for Question 22 is 3 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 3

TOPIC: **Probability Toolkit**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Probability Toolkit. The tag is correct.

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FINAL ANSWER

$\frac{51}{76}$

PAST PAPER QUESTION

Q#: 23 Marks: 6

TOPIC: **3D Pythagoras & Trigonometry**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

23 The diagram shows cuboid *ABCDEFGH*

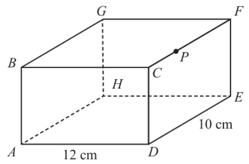

Diagram **NOT** accurately drawn

The point P lies on CF

$AD = 12$ cm $\quad DE = 10$ cm $\quad CP = 3$ cm

The angle of elevation of P from A is $24°$

Calculate the size of angle APG
Give your answer correct to the nearest degree.
Show your working clearly.

.... °

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 6

TOPIC: **3D Pythagoras & Trigonometry**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Was tagged Algebra Toolkit; actually 3D Pythagoras & Trigonometry.

Solution

Point P is 3 cm from C along CF. The horizontal distance from A to the point directly below P is

$$\sqrt{12^2 + 3^2} = \sqrt{153}$$

Using the angle of elevation,

$$h = \sqrt{153}\tan 24^\circ = 5.507\ldots$$

So

$$AP = \frac{\sqrt{153}}{\cos 24^\circ} = 13.5399\ldots$$

On the top face,

$$PG = \sqrt{12^2 + 7^2} = \sqrt{193} = 13.8924\ldots$$

Also,

$$AG = \sqrt{12^2 + 10^2 + h^2} = 16.561\ldots$$

Use the cosine rule in triangle APG.

$$\cos\angle APG = \frac{AP^2 + PG^2 - AG^2}{2(AP)(PG)}$$

$$\angle APG = 49.163\ldots^\circ$$

To the nearest degree,

$$49^\circ$$

FINAL ANSWER

49°

PAST PAPER QUESTION Q#: 23 Marks: 6

Topic: **3D Pythagoras & Trigonometry**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

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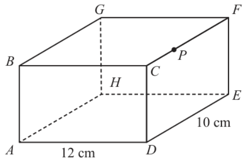

Diagram **NOT** accurately drawn

The point *P* lies on *CF*

$AD = 12$ cm $DE = 10$ cm $CP = 3$ cm

The angle of elevation of *P* from *A* is 24°

Calculate the size of angle *APG*
Give your answer correct to the nearest degree.
Show your working clearly.

.... °

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 6

TOPIC: **3D Pythagoras & Trigonometry**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

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$$\angle APG = 49.163\ldots^\circ$$

To the nearest degree,

$$49^\circ$$

FINAL ANSWER

49°

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Coordinate Geometry**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

24 *PQRS* is a square.

PR is a diagonal of the square.

P is the point with coordinates (4, 7)
R is the point with coordinates (8, –5)

Find an equation of the straight line that passes through the points *Q* and *S*
Give your answer in the form $ay = bx + c$ where a, b and c are integers.

....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Coordinate Geometry**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Was tagged Algebra Toolkit; actually Linear Graphs.

Solution

The diagonals of a square bisect each other and are perpendicular.
Midpoint of $P(4,7)$ and $R(8,-5)$:

$$\left(\frac{4+8}{2}, \frac{7-5}{2}\right) = (6,1)$$

Gradient of PR:

$$\frac{-5-7}{8-4} = -3$$

So the perpendicular gradient is $\frac{1}{3}$.
The line through Q and S passes through $(6,1)$, so

$$y - 1 = \frac{1}{3}(x-6)$$

$$3y - 3 = x - 6$$

$$3y = x - 3$$

FINAL ANSWER

$3y = x - 3$

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Coordinate Geometry**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

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Give your answer in the form $ay = bx + c$ where a, b and c are integers.

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(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

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May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

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The line through Q and S passes through $(6,1)$, so

$$y - 1 = \frac{1}{3}(x-6)$$

$$3y - 3 = x - 6$$

$$3y = x - 3$$

FINAL ANSWER
$3y = x - 3$

PAST PAPER QUESTION Q#: 25 Marks: 5

TOPIC: **Completing the Square**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

25 (a) Express $8 - 12x - 4x^2$ in the form $a - b(x + c)^2$
where a, b and c are constants.

....

(3)

(b) Hence solve $8 - 12x - 4x^2 = 0$
Show your working clearly.
Give your solutions in surd form.

....

(2)

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 5

TOPIC: **Completing the Square**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Was tagged Surds; actually Completing the Square.

Solution

$$8x^2 - 12x - 4 = 8\left(x^2 - \frac{3}{2}x\right) - 4$$

Complete the square.

$$x^2 - \frac{3}{2}x = \left(x - \frac{3}{4}\right)^2 - \frac{9}{16}$$

So

$$8x^2 - 12x - 4 = 8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2}$$

Now solve

$$8x^2 - 12x - 4 = 0$$

$$8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2} = 0$$

$$\left(x - \frac{3}{4}\right)^2 = \frac{17}{16}$$

$$x - \frac{3}{4} = \pm\frac{\sqrt{17}}{4}$$

$$x = \frac{3 \pm \sqrt{17}}{4}$$

FINAL ANSWER

$8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2}$, and $x = \frac{3\pm\sqrt{17}}{4}$.

PAST PAPER QUESTION

Q#: 25 **Marks: 5**

TOPIC: **Completing the Square**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

25 (a) Express $8 - 12x - 4x^2$ in the form $a - b(x + c)^2$
where a, b and c are constants.

....
(3)

(b) Hence solve $8 - 12x - 4x^2 = 0$
Show your working clearly.
Give your solutions in surd form.

....
(2)

(Total for Question 25 is 5 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 5

Topic: **Completing the Square**

May 2025 4WM1H - May2025 | 4WM1H | Unit 1

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Was tagged Surds; actually Completing the Square.

Solution

$$8x^2 - 12x - 4 = 8\left(x^2 - \frac{3}{2}x\right) - 4$$

Complete the square.

$$x^2 - \frac{3}{2}x = \left(x - \frac{3}{4}\right)^2 - \frac{9}{16}$$

So

$$8x^2 - 12x - 4 = 8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2}$$

Now solve

$$8x^2 - 12x - 4 = 0$$

$$8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2} = 0$$

$$\left(x - \frac{3}{4}\right)^2 = \frac{17}{16}$$

$$x - \frac{3}{4} = \pm\frac{\sqrt{17}}{4}$$

$$x = \frac{3 \pm \sqrt{17}}{4}$$

FINAL ANSWER

$8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2}$, and $x = \frac{3\pm\sqrt{17}}{4}$.